

Total No. of Questions : 8]

SEAT No. :

PA-1500

[Total No. of Pages : 3

[5926]:120

T.E. (Information Technology)
OPERATING SYSTEMS
(2019 Pattern) (Semester-I) (314442)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3. or Q.4, Q.5 or Q.6, and Q.7 or Q.8.
- 2) Figures to the right side indicate full marks.

- Q1)** a) What is semaphore and mutex? Explain with the help of pseudocode, how semaphore is used to solve producer consumer problem? [9]
- b) What are the four necessary conditions for deadlock? How is a deadlock detected in a system with resources having single instances? Explain with an example. [9]

OR

- Q2)** a) Define mutual exclusion, race condition, semaphore. Deadlock. [6]
- b) What is Bankers safe sequence algorithm? Apply it for finding safe sequence of execution of 5 processes in a system having Snapshot at time T₀: [12]

	Allocation	Max	Available
	A B C	A B C	A B C
P0	0 1 0	7 5 3	3 3 2
P1	2 0 0	3 2 2	
P2	3 0 2	9 0 2	
P3	2 1 1	2 2 2	
P4	0 0 2	4 3 3	

Also determine whether following requests can be granted or not:

- i) Request for process P2: - 3 0 0 and
- ii) Request for process P3: - 0 0 1

P.T.O.

- Q3) a)** Given memory partitions of 150k, 650k, 280k, 390k and 540k (in order) how would each of the First fit, Best fit, and Worst fit algorithms place processes of 212k, 457k, 112k, 510k and 326k (in order) [9]
- b) With the help of neat diagrams, Write a short note on Buddy system. [8]

OR

- Q4) a)** Explain Belady's anomaly with suitable example. [6]
- b) Consider the following segment table: [6]

Segment	Base	Length
0	1790	350
1	2722	1050
2	520	925
3	5200	450
4	4200	655

What are the physical addresses for the following logical addresses?

- i) 0,330
 - ii) 2,525
 - iii) 4,700
 - iv) 3,400
 - v) 1,1110
- c) What are the distinctions among logical, relative and physical addresses? [5]

- Q5) a)** A disk drive has 200 tracks, numbered 0-199. The drive is currently serving the request at track no 53. The queue of pending requests in FIFO order is 98, 183, 37, 122, 14, 124, 65, 67. Starting from the current head position what is the total distance that disk arm moves to satisfy all the pending requests for the following disk scheduling algorithms. Assume that the head is moving in the increasing order of track number for SCAN and C-LOOK. [12]

- i) FCFS
 - ii) SCAN
 - iii) C-LOOK
 - iv) SSTF
- b) Explain with diagrams different I/O buffering techniques. [6]

OR

- Q6)** a) List and explain different file access methods. [9]
- b) Describe different methods of record blocking with the help of a neat diagram. [9]
- Q7)** a) What is system software? Explain any four system software in brief? [6]
- b) Explain imperative statement, declarative statement, and assembly directive of assembly language programming? [6]
- c) Discuss with example what is forward reference problem. [5]

OR

- Q8)** a) Explain ORIGIN, EQU and LTROG with an example. [6]
- b) Explain the data structures required for two PASS Assembler in detail. [6]
- c) Differentiate between literal and immediate operand. [5]



Total No. of Questions : 8]

SEAT No. :

P483

[Total No. of Pages : 2

[6003]:702

T.E. (Information Technology)
OPERATING SYSTEMS
(2019 Pattern) (Semester-I) (314442)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) What conditions are generally associated with readers-writers problem?
Write its pseudo code. **[9]**

b) Describe resource allocation graph in detail. **[9]**

OR

Q2) a) Enlist different IPC techniques. Differentiate between named pipe and unnamed pipe with suitable example **[9]**

b) What is Critical Section Problem? Give semaphore solution for producer-consumer problem. **[9]**

Q3) a) Consider six memory partitions of size 100 KB, 300 KB, 50 KB, 200 KB, 150 KB and 200 KB. These partitions need to be allocated to processes of sizes 200 KB, 100 KB, 50 KB in that order. Perform the allocation of processes using dynamic partitioning algorithms given below and comment on internal and external fragmentation- **[12]**

- i) First Fit Algorithm
- ii) Best Fit Algorithm
- iii) Worst Fit Algorithm

b) Explain Buddy system memory allocation with suitable example. **[5]**

OR

P.T.O.

- Q4) a)** Find the number of page faults for the reference string 6,5,1,2,5,3,5,4,2,3,6,3,2,1,2 using FIFO, LRU and optimal page replacement strategies. Consider frame size as 3. [12]
- b)** Explain Belady's anomaly with suitable example. [5]

- Q5) a)** Assume a disk with 200 tracks and the disk request queue has random requests in it as follows: 55,58,39,18,90,160,150,38,184. Find the no of tracks traversed and average seek length if

- 1) SSTF
- 2) SCAN
- 3) C-SCAN

Is used and initially head is at track no 100. [12]

- b)** What are typical operations that may be performed on a directory? [6]

OR

- Q6) a)** What is I/O buffering? Why I/O buffering is needed? State and explain different approaches of I/O buffering. [9]

- b)** Explain with example any three disk scheduling criteria. [9]

- Q7) a)** List down the phases of a compiler. Explain with suitable example. [9]

- b)** Explain macro call and macro expansion with suitable example. [8]

OR

- Q8) a)** Explain with example imperative statement, declarative statement, and assembly directive of assembly language programming? [9]

- b)** What is system software explain any four system software in brief? [8]



Total No. of Questions : 8]

SEAT No. :

P-7618

[Total No. of Pages : 2

[6180]-138

T.E. (Information Technology)

OPERATING SYSTEMS

(2019 Pattern) (Semester-I) (314442)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

- Q1)** a) Describe in brief the different IPC mechanisms. [9]
b) What is deadlock? Explain the necessary and sufficient conditions for the occurrence of a deadlock. [9]

OR

- Q2)** a) Explain the following with example [9]
i) Mutual Exclusion ii) Synchronization
iii) Race condition.
b) What is Critical Section Problem? Explain readers-writers problem. [9]

- Q3)** a) What is page Fault? For the given reference string with 3-page frame available, determine the number of page faults for FIFO and LRU algorithms: [9]
3, 5, 3, 7, 2, 1, 5, 4, 6, 7, 4, 1, 2.
b) Explain Demand paging with the help of neat diagram. [8]

OR

- Q4)** a) Explain Buddy System with the help of neat diagram and example. [9]
b) What is segmentation? How address Translation is performed in segmentation system? [8]

P.T.O.

- Q5) a)** Describe three methods of record blocking with the help of neat diagrams. [9]
- b)** Explain the different functions of a file management system. [9]

OR

- Q6) a)** Assume a disk with 200 tracks and the disk request queue has random Requests in it as follows: 98,183,37,122,14,124,65,67. [12]

Find the no of tracks traversed and average seek length if [6]

- i) FCFS
 - ii) SSTF
 - iii) SCAN is used and initially head is at track no 53.
- b)** What is spooling? Explain with suitable diagrams. [6]

- Q7) a)** Explain the data structures required for two PASS Assembler in detail. [6]
- b)** What is Macro? Explain macro call and macro expansion with suitable example. [6]
- c)** Discuss with example what is forward reference problem. [5]

OR

- Q8) a)** What is Loader? What are the basic functions of loaders? [10]
- b)** What is system software? Explain any four system software in brief ? [7]



Total No. of Questions : 8]

SEAT No. :

PB3859

[6262]-122

[Total No. of Pages : 2

**T.E. (Information Technology)
OPERATING SYSTEMS
(2019 Pattern) (Semester-I) (314442)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume Suitable data if necessary.

Q1) a) Explain the following with example **[9]**

- i) Mutual Exclusion
- ii) Synchronization
- iii) Race condition

b) What is Critical Section problem? Explain readers- writers problem. **[9]**

OR

Q2) a) Enlist different IPC techniques. Differentiate between named pipe and unnamed pipe with suitable example. **[9]**

b) What is Critical Section problem? Give semaphore solution for producer-consumer problem. **[9]**

Q3) a) What is page Fault? For the given reference string with 3- page frame available, determine the number of page faults for FIFO and LRU algorithms: 3,5,7,2,1,5,4,6,7,4,1,2. **[9]**

b) Explain Demand paging with the help of neat diagram. **[8]**

OR

Q4) a) Explain Buddy System with the help of neat diagram and example. **[9]**

b) What is segmentation? How address Translation is performed in segmentation system? **[8]**

P.T.O.

- Q5) a)** Assume a disk with 200 tracks and the disk request queue has random requests in it as follows: 55,58,39,18,90,160,150,38,184. [12]

Find the no of tracks traversed and average seek length if

- i) SSTF
 - ii) SCAN
 - iii) C-SCAN is used and initially head is at track no 100.
- b) What are typical operations that may be performed on a directory? [6]

OR

- Q6) a)** What is I/O buffering? Why I/O buffering is needed? State and explain different approaches of I/O buffering. [9]
- b) Explain with example any three disk scheduling criteria. [9]

- Q7) a)** List down the phases of a compiler. Explain with suitable example. [9]
- b) Explain macro call and macro expansion with suitable example. [8]

OR

- Q8) a)** What is Loader? What are the basic functions of loaders? [10]
- b) What is system software? Explain any four system software in brief?[7]



Total No. of Questions : 8]

SEAT No. :

PC1802

[Total No. of Pages : 2

[6353] 121

T.E. (Information Technology)

OPERATING SYSTEMS

FMG - 6 : Financial Mathematics

(2019 Pattern) (Semester - I) (314442)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Assume Suitable data if necessary.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Figures to the right side indicate full marks.

Q1) a) State the dining philosopher problem. Provide solution by using semaphores and discuss how the critical section requirements are fulfilled? [9]

b) Describe methods for deadlock recovery. [8]

OR

Q2) a) Consider the following snapshot of the system [10]

	Allocation	Max	Available
	R ₁ R ₂ R ₃	R ₁ R ₂ R ₃	R ₁ R ₂ R ₃
P ₁	0 1 0	7 5 3	2 3 0
P ₂	3 0 2	3 2 2	
P ₃	3 0 2	9 0 2	
P ₄	2 1 1	2 2 2	
P ₅	0 0 2	4 3 3	

Answer the following questions using Banker's Algorithm

- i) What are the values of need matrix?
 - ii) Is the system in the safe mode? If yes, what is the safe sequence?
 - iii) Request for process P1-> 2 0 1, determine whether the request can be granted safely or not? if yes then write safe sequence.
- b) Define Mutual Exclusion. List the requirements of mutual exclusion.[7]

P.T.O.

- Q3) a)** For the following reference string. [12]
 1, 2, 3, 4, 5, 3, 4, 1, 6, 7, 8, 7, 8, 9, 5, 4, 2, 4, 9
 Count the number of page faults that occur with 3 frames using FIFO, Optimal and LRU page replacement methods. Discuss the result.
- b)** Write short note on virtual memory management. [6]
- OR
- Q4) a)** Describe dynamic partitioning memory allocation strategies with example [10]
- b)** Elaborate the concept of demand paging with appropriate diagram [8]
- Q5) a)** A disk drive has 200 cylinders, numbered 0-199. The drive is currently serving the request at cylinder 53. The queue of pending requests in FIFO order is 98, 183, 37, 122, 14, 124, 65, 67. Starting from the current head position, what is the total distance that the disk arms moves to satisfy all the pending requests for the following disk scheduling algorithms. Assume head is moving towards end of disk for C-SCAN and LOOK. [12]
- i) FCFS
 ii) C-SCAN
 iii) LOOK
 iv) SSTF
- b)** State and explain different I/O buffering techniques. [6]
- OR
- Q6) a)** Explain different file allocation techniques. [8]
- b)** Write a short note on disk free space management. [6]
- c)** Describe File sharing. [4]
- Q7) a)** Draw a general model of compiler and explain all phases in detail. [9]
- b)** What are the types of loaders? Discuss four different functions of loaders. [8]
- OR
- Q8) a)** Define Lexeme, Token and Pattern. [6]
- b)** Discuss forward reference problem with example. [5]
- c)** Explain the 'Compile and Go' Loader scheme with advantages and disadvantages using a suitable diagram. [6]

Total No. of Questions : 8]

SEAT No. :

PD4323

[Total No. of Pages : 2

[6403]-121

T.E. (Information Technology)
OPERATING SYSTEMS
(2019 Pattern) (Semester - V) (314442)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume suitable data wherever necessary.

- Q1)** a) Explain the Reader writer problem with the readers having priority and give a solution using semaphore and mutex. [8]
b) Explain various strategies to deal with deadlocks. [9]

OR

- Q2)** a) State the Producer-Consumer problem and give a solution using semaphore and mutex. [7]
b) Define and discuss : [10]
i) Mutex
ii) Race condition
iii) Critical section
iv) Semaphore
v) Deadlock

- Q3)** a) For the reference string: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 3, 6. Calculate the number of page faults with 3 frames using FIFO, LRU, and optimal page replacement methods. [9]
b) Differentiate : [9]
i) Logical and physical address
ii) Page table and segment table
iii) Static and dynamic partitioning

OR

- Q4)** a) Explain demand paging. Why do we need demand paging? [6]
b) Explain thrashing with an example. How can it be handled? [6]
c) Discuss the Internal and External Fragmentation with example. [6]

P.T.O.

Q5) a) Assume a disk with 400 tracks, and the disk request has random requests as follows: 53, 59, 42, 21, 310, 350, 380, 184. Find the no. of tracks traversed and average seek length if [9]

i) FIFO

ii) SSTF

iii) CLOOK is used, and initially head is at track no.100.

b) Why I/O buffering is necessary? State and explain different I/O buffering techniques. [8]

OR

Q6) a) Why free space management is required. Explain various Free space management techniques (at least 4) [9]

b) Explain different file organization techniques with their advantages and disadvantages. [8]

Q7) a) Categorize the overall compilation process in the form of various phases using suitable diagrams. Explain each phase in detail. [9]

b) Explain the data structures required for a two-pass assembler with their usage in detail. [9]

OR

Q8) a) Define System Software. How is it different from application software? List and explain important components of system software. [9]

b) Explain the following : [9]

i) Compile and Go Loader

ii) Direct linking loader

